Any two elements in the first row of T must be in different columns of T', so there exists $q_1' \in Q_\lambda'$ which moves all these elements to the first row. So there is $p_1 \in P_\lambda$ such that $p_1 T$ and $q_1' T'$ have the same first row. Now do the same procedure with the second row, finding elements p_2, q_2' such that $p_2 p_1 T$ and $q_2' q_1' T'$ have the same first two rows. Continuing so, we will construct the desired elements p, q'. The lemma is proved. □

Let us introduce the **lexicographic ordering** on partitions: $\lambda > \mu$ if the first nonvanishing $\lambda_i - \mu_i$ is positive.

Lemma 5.13.2. *If $\lambda > \mu$, then $a_\lambda \mathbb{C}[S_n] b_\mu = 0$.*

Proof. Similarly to the previous lemma, it suffices to show that for any $g \in S_n$ there exists a transposition $t \in P_\lambda$ such that $g^{-1} t g \in Q_\mu$. Let $T = T_\lambda$ and $T' = g T_\mu$. We claim that there are two integers which are in the same row of T and the same column of T'. Indeed, if $\lambda_1 > \mu_1$, this is clear by the pigeonhole principle (already for the first row). Otherwise, if $\lambda_1 = \mu_1$, as in the proof of the previous lemma, we can find elements $p_1 \in P_\lambda, q_1' \in g Q_\mu g^{-1}$ such that $p_1 T$ and $q_1' T'$ have the same first row and repeat the argument for the second row, and so on. Eventually, having done $i-1$ such steps, we'll have $\lambda_i > \mu_i$, which means that some two elements of the ith row of the first tableau are in the same column of the second tableau, completing the proof. □

Lemma 5.13.3. *c_λ is proportional to an idempotent. Namely, $c_\lambda^2 = \frac{n!}{|P_\lambda||Q_\lambda| \dim V_\lambda} c_\lambda$.*

Proof. Lemma 5.13.1 implies that c_λ^2 is proportional to c_λ. Also, it is easy to see that the trace of c_λ in the regular representation is $n! |P_\lambda|^{-1} |Q_\lambda|^{-1}$ (as the coefficient of the identity element in c_λ is $|P_\lambda|^{-1} |Q_\lambda|^{-1}$). This implies the statement. □

Lemma 5.13.4. *Let A be an algebra and let e be an idempotent in A. Then for any left A-module M, one has $\mathrm{Hom}_A(Ae, M) \cong eM$ (namely, $x \in eM$ corresponds to $f_x : Ae \to M$ given by $f_x(a) = ax$, $a \in Ae$).*